Don't Look Back in Anger: Wasserstein Distributionally Robust Optimization with Nonstationary Data

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Abstract: We study data-driven decision problems where historical observations are generated by a time-evolving distribution whose consecutive shifts are bounded in Wasserstein distance. We address this nonstationarity using a distributionally robust optimization model with an ambiguity set that is a Wasserstein ball centered at a *weighted* empirical distribution, thereby allowing for the time decay of past data in a way which accounts for the drift of the data-generating distribution. Our main technical contribution is a concentration inequality for weighted empirical distributions that explicitly captures both the effective sample size (i.e., the equivalent number of equally weighted observations) and the distributional drift. Using our concentration inequality, we select observation weights that optimally balance the effective sample size against the extent of drift. The family of optimal weightings reveals an interplay between the order of the Wasserstein ambiguity ball and the time-decay profile of the optimal weights. Classical weighting schemes, such as time windowing and exponential smoothing, emerge as special cases of our framework, for which we derive principled choices of the parameters. Numerical experiments demonstrate the effectiveness of the proposed approach.

1 Introduction

Many practical decision problems involve problem data that are uncertain at the point of decision. Examples include inventory control with uncertain demands, maintenance scheduling under uncertain failures, online advertising with uncertain click-through rates, and portfolio allocation under uncertain market conditions.

Classical stochastic programming models the uncertain quantities as random variables and optimizes a risk measure – such as the expectation or (conditional) value-at-risk – under a known data-generating distribution. In practice, however, this distribution is rarely known. Data-driven approaches therefore replace the unknown distribution with an estimate formed from historical observations. The most prominent example is the sample average approximation (SAA), which optimizes against the empirical distribution that assigns equal mass to past samples. While computationally appealing and asymptotically optimal, SAA is prone to overfitting in small- to moderate-sample regimes – a phenomenon often described as the "optimizer's curse." Distributionally robust optimization (DRO) mitigates overfitting by optimizing against the worst distribution in an ambiguity set that contains all distributions deemed plausible given the data. Among different ambiguity models, Wasserstein balls around an empirical reference measure have emerged as a common choice since they respect the geometry of the sample space, yield finite-sample performance guarantees, and often lead to tractable formulations.

The vast majority of research on data-driven optimization presumes that observations are independent and identically distributed over time. In practice, this assumption is often untenable: consumer preferences shift, supply conditions change, and competitors, whose actions are often modeled as exogenous uncertainty, adapt over time. Under nonstationarity, pooling all past observations with equal weight – as in standard SAA and Wasserstein DRO – can be harmful: looking at a long historical sequence of observations can increase statistical precision but may bias estimates toward outdated regimes, whereas focusing only on recent data can reduce bias but may inflate variance.

This paper proposes a Wasserstein DRO framework tailored to nonstationary data. We model the evolution of the data-generating process as a sequence of unknown distributions whose successive changes are bounded in Wasserstein distance. To derive robust decisions, we center our ambiguity sets at *weighted* empirical distributions that downweight older samples. We then consider the fundamental variance–drift trade-off that governs learning under nonstationarity. More specifically, our contributions are as follows.

(i) **Modeling nonstationarity via Wasserstein shifts.** We propose a distributionally robust model where the data-generating distribution changes over time, with consecutive shifts bounded in Wasserstein distance. Our ambiguity sets are centered at weighted empirical measures, and they enable principled time decay of historical information within DRO.

(ii) **Weighted concentration for Wasserstein distances.** We extend the classical concentration inequalities of [7] from equally weighted to weighted empirical distributions and obtain finite-sample guarantees for nonstationary data, with rates that explicitly reflect both the effective sample size and the magnitude of distributional drift.

(iii) **Optimizing the variance–drift trade-off.** We show how to choose weights that optimally balance statistical variance against distributional drift. Our analysis covers natural weight families (e.g., time windowing and exponential decay) and characterizes optimal weights within our framework.

Taken together, these contributions yield a conceptually simple and computationally practical method for decision-making with nonstationary data: center a Wasserstein ambiguity set at a weighted empirical distribution, set the weights – guided by our weighted concentration bounds – to balance variance and drift, and calibrate the ambiguity radius via cross-validation.

Our work intersects with, and contributes to, two strands of the literature: DRO under nonstationary data-generating processes and DRO with heterogeneous data sources.

The literature on DRO with nonstationary data relaxes the ubiquitous identically distributed assumption by modeling temporal dependence or drift. For example, [2, 11, 19] study Wasserstein DRO models where data are generated by vector autoregressive processes with unknown parameters. Likewise, [14, 19] analyze Wasserstein DRO models where data arise from unknown finite-state Markov chains. In addition, [16] models nonstationarity via a regime-switching Wasserstein ambiguity set, and [3] investigates φ-divergence DRO where data are generated by certain classes of Markov chains. A common feature of this literature is the imposition of parametric or Markovian structure on the temporal dynamics. Instead, we only assume that successive data-generating distributions differ in Wasserstein distance by no more than a fixed amount, without committing to a specific parametric time-series model.

In contrast, the literature on DRO with heterogeneous data sources leverages samples from multiple distributions (sources) that are close to the data-generating (target) distribution in Wasserstein distance to construct an ambiguity set for the target. Focusing on two sources and least squares estimation, [20] proposes either interpolating between the ambiguity sets centered at each source or intersecting them. For logistic regression with two sources that are subjected to adversarial attacks, [18] intersects the two ambiguity sets. In the first work that studies more than two sources, [17] intersects Wasserstein balls centered at all sources and optimizes against the worst distribution in the intersection. The paper shows that the problem is NP-hard in general but can be solved in polynomial time when either the number of sources or the dimension of the uncertain parameters is fixed. Finally, [9] studies a dynamic setting where source reliability varies over time and proposes mechanisms to update trust in each source. The literature on DRO with heterogeneous data sources typically assumes that many samples are available per source. On the other hand, viewing each time step as a separate source, our setting provides exactly one observation per source. We will see that this renders the existing heterogeneous-source guarantees unsuited. Instead, our approach aggregates over time via weighted empiricals while explicitly controlling for the per-step Wasserstein drift.

The remainder of the paper is organized as follows. Section 2 introduces our framework. Section 3 develops new concentration inequalities for weighted empirical distributions. Section 4 characterizes weight choices that optimally balance variance and drift and derives optimal tuning rules for classical schemes such as time windowing and exponential smoothing. Section 5 reports numerical results, and Section 6 concludes. All source code and results are available in the GitHub repository accompanying this work.[1]

Notation. $\|\cdot\|$ denotes the standard Euclidean norm. $\mathfrak{P}(\Xi)$ denotes the set of all Borel probability measures on a set Ξ. Among these measures, the point-mass distribution $\mathbb{1}_\xi$ assigns probability 1 to the outcome $\xi \in \Xi$. For a distribution $\mathbb{P} \in \mathfrak{P}(\Xi)$, we write $\boldsymbol{\xi} \sim \mathbb{P}$ to express that a random vector $\boldsymbol{\xi}$ is distributed according to $\mathbb{P}$. Boldface symbols denote random vectors, and plain symbols denote their outcomes. $\mathcal{W}_N$ denotes the probability N-simplex; that is, the set of non-negative N-tuples summing to 1. Also, we write $[T] \coloneqq \{1, \ldots, T\}$ and $x_+ \coloneqq \max\{0, x\}$.

2 Problem Setting

We study data-driven decision problems where a decision maker observes T historical samples $\boldsymbol{\xi}_1 \sim \mathbb{P}_1, \ldots, \boldsymbol{\xi}_T \sim \mathbb{P}_T$ of an uncertain parameter $\xi \in \Xi$ governed by unknown time-varying data-generating distributions $\mathbb{P}_1, \ldots, \mathbb{P}_T \in \mathfrak{P}(\Xi)$ supported on the closed set $\Xi \subset \mathbb{R}^m$. At time $T+1$, the decision maker selects a measurable, extended real–valued loss function $\ell : \Xi \to \mathbb{R} \cup \{\infty\}$ from the admissible class $\mathcal{L}$ to solve

$$\underset{\ell \in \mathcal{L}}{\text{minimize}} \quad \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}}\big[\ell(\boldsymbol{\xi})\big], \tag{1}$$

[1] Link: `https://github.com/dominickeehan/dont-look-back-in-anger`.

where $\mathcal{P} \subset \mathfrak{P}(\Xi)$ is the ambiguity set for the (unknown) next-period distribution $\mathbb{P}_{T+1}$. In the special case where losses are generated by a cost function $f : \mathcal{X} \times \Xi \to \mathbb{R} \cup \{\infty\}$ depending on both a decision variable $x \in \mathcal{X}$ and an uncertain variable ξ, the admissible class is $\mathcal{L} = \{\xi \mapsto f(x, \xi) : x \in \mathcal{X}\}$.

When $\mathbb{P}_1 = \cdots = \mathbb{P}_{T+1}$ and $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_T$ are drawn independently (the *stationary* setting), a common choice for the ambiguity set in (1) is $\mathcal{P} = \mathbb{B}_p(\frac{1}{T}\sum_{t=1}^T \mathbb{1}_{\boldsymbol{\xi}_t}; \varepsilon)$; the p-Wasserstein ball of radius ε centered around the empirical distribution $\frac{1}{T}\sum_{t=1}^T \mathbb{1}_{\boldsymbol{\xi}_t}$. Here

$$\mathbb{B}_p(\mathbb{P}; \varepsilon) \coloneqq \Big\{\mathbb{Q} \in \mathfrak{P}(\Xi) : W_p(\mathbb{P}, \mathbb{Q}) \leq \varepsilon\Big\}, \quad W_p(\mathbb{P}, \mathbb{Q}) \coloneqq \Big(\inf_{\gamma \in \Gamma(\mathbb{P}, \mathbb{Q})} \mathbb{E}_\gamma\big[\|\boldsymbol{\xi} - \boldsymbol{\zeta}\|^p\big]\Big)^{1/p},$$

and $\Gamma(\mathbb{P}, \mathbb{Q})$ is the set of couplings of $\mathbb{P}$ and $\mathbb{Q}$. This choice is popular because it yields tractable reformulations of the problem (1) for broad classes of loss functions and, under mild conditions, satisfies a finite-sample guarantee of the form

$$\Pr\Big[\mathbb{P}_{T+1} \notin \mathbb{B}_p\Big(\tfrac{1}{T}\textstyle\sum_{t=1}^T \mathbb{1}_{\boldsymbol{\xi}_t}; \varepsilon\Big)\Big] \leq c_1 \cdot \exp\Big(-c_2 T \varepsilon^m\Big)$$

for positive constants c_1 and c_2 that depend only on Ξ and p. Thus, the unknown true distribution is contained in $\mathbb{B}_p(\frac{1}{T}\sum_{t=1}^T \mathbb{1}_{\boldsymbol{\xi}_t}; \varepsilon)$ with high, controllable, probability.

We depart from the stationary setting by allowing $\mathbb{P}_t$ to vary over time, but we assume that consecutive changes are bounded in Wasserstein distance. In other words, we stipulate that $W_\infty(\mathbb{P}_t, \mathbb{P}_{t+1}) \leq \rho$ for all $t \in [T]$ and some non-negative drift bound ρ, where

$$W_\infty(\mathbb{P}, \mathbb{Q}) \coloneqq \lim_{p \to \infty} W_p(\mathbb{P}, \mathbb{Q}) = \inf_{\gamma \in \Gamma(\mathbb{P}, \mathbb{Q})} \operatorname*{ess-sup}_{(\boldsymbol{\xi}, \boldsymbol{\zeta}) \sim \gamma} \|\boldsymbol{\xi} - \boldsymbol{\zeta}\|.$$

We continue to assume that, given the distributions $\mathbb{P}_1, \ldots, \mathbb{P}_T$, the samples $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_T$ are independent. Bounding the distributional drift via W_∞ allows us to control the worst-case pointwise displacement between consecutive distributions, and it implies that $W_p(\mathbb{P}_t, \mathbb{P}_{t+1}) \leq \rho$ for all $p \in [1, \infty]$.

In the presence of a positive drift ρ, the Wasserstein ambiguity set $\mathbb{B}_p(\frac{1}{T}\sum_{t=1}^T \mathbb{1}_{\boldsymbol{\xi}_t}; \varepsilon)$ centered around the empirical distribution $\frac{1}{T}\sum_{t=1}^T \mathbb{1}_{\boldsymbol{\xi}_t}$ overweights earlier observations that are less representative of the next-period distribution $\mathbb{P}_{T+1}$. A seemingly natural alternative, inspired by the literature on DRO with heterogeneous sources, is to take the intersection of balls around each single-sample "data source" estimate $\mathbb{1}_{\boldsymbol{\xi}_t}$, that is, $\mathcal{P} = \bigcap_{t=1}^T \mathbb{B}_p(\mathbb{1}_{\boldsymbol{\xi}_t}; \varepsilon_t)$ for some radii $\varepsilon_1, \ldots, \varepsilon_T$, where ε_t is chosen to reflect that $\mathbb{P}_{T+1}$ is at a greater distance from earlier distributions. We next show that this construction may exhibit poor large-sample behavior.

Example 1 (Asymptotic Inconsistency of the Intersection Ambiguity Set)**.** For $\beta \in [0, 1)$, let $\varepsilon(\beta)$ denote the $(1 - \beta)$-confidence p-Wasserstein radius (in the stationary setting) around an empirical distribution based on a single sample (as may be obtained from [7, Theorem 2]). If the confidence radius is set from a concentration bound, then $\varepsilon(\,\cdot\,)$ is nonincreasing and strictly positive (see also, e.g., [6]). In our time-varying setting with drift bound ρ, choose violation levels $\beta_1, \ldots, \beta_T \in [0, 1)$ such that $\sum_{t=1}^T \beta_t = \beta \in [0, 1)$ and set

$$\varepsilon_t \coloneqq \varepsilon(\beta_t) + (T - t + 1)\rho \quad \text{for all } t \in [T],$$

which achieves the guarantee [17, Proposition 4]

$$\Pr\left[\mathbb{P}_{T+1} \notin \bigcap_{t=1}^{T} \mathbb{B}_p\Big(\mathbb{1}_{\xi_t}; \varepsilon(\beta_t) + (T-t+1)\rho\Big)\right] \leq \sum_{t=1}^{T} \beta_t = \beta. \tag{2}$$

We claim that this confidence guarantee is unsatisfactory when the drift bound is small and the sample size is large. In particular, consider the case where the data-generating process is actually stationary with $\mathbb{P}_1 = \cdots = \mathbb{P}_{T+1} = \mathbb{1}_\xi$ for some $\xi \in \Xi$. Then each $\boldsymbol{\xi}_t = \xi$, so all of the balls in (2) are centered around $\mathbb{1}_\xi$ and satisfy

$$\bigcap_{t=1}^{T} \mathbb{B}_p\Big(\mathbb{1}_\xi; \varepsilon(\beta_t) + (T-t+1)\rho\Big) = \mathbb{B}_p\Big(\mathbb{1}_\xi; \min_{t\in[T]}\big\{\varepsilon(\beta_t) + (T-t+1)\rho\big\}\Big).$$

Since $\beta_t \leq \sum_{s=1}^{T} \beta_s = \beta$, the monotonicity and strict positivity of $\varepsilon(\,\cdot\,)$ imply that

$$\min_{t\in[T]}\big\{\varepsilon(\beta_t) + (T-t+1)\rho\big\} \geq \varepsilon(\beta) > 0,$$

and thus

$$\bigcap_{t=1}^{T} \mathbb{B}_p\Big(\mathbb{1}_\xi; \varepsilon(\beta_t) + (T-t+1)\rho\Big) \supset \mathbb{B}_p(\mathbb{1}_\xi; \varepsilon(\beta)) \neq \{\mathbb{1}_\xi\}.$$

Hence, with $\rho = 0$, even as $T \to \infty$ the intersection ambiguity set cannot collapse to $\mathbb{1}_\xi$. This is because each constituent ball is centered at a single-point empirical distribution, so the confidence radii do not decrease with T. □

The failure of the intersection-based ambiguity set to contract with T – even in the absence of drift – motivates us to center ambiguity sets at *weighted* empirical distributions $\sum_{t=1}^{T} w_t \mathbb{1}_{\boldsymbol{\xi}_t}$. The next section develops finite-sample concentration inequalities for such weighted empirical distributions under bounded drift, and Section 4 leverages these results to optimize the weights.

We compare the intersection approach to the weighted approach, that we discuss in following sections, using a one-dimensional example with $p = 2$, $\rho = 1/3$, and observations $\xi_1 = 1$, $\xi_2 = -1$, $\xi_3 = 2$, and $\xi_4 = 3$. For the weighted approach we set $\varepsilon = 3$ and for the intersection approach we scale this up by a factor of 1.5 so that the eventual ambiguity sets are visually comparable. (Our aim with this example is not to compare the associated confidence guarantees, but the geometry of each set; the associated guarantees are *not* comparable from this example and will differ here even before scaling.) Figure 1 shows which uniform distributions lie in each set. The figure reports the ranges of means and standard deviations of the distributions. The weighted approach uses $w_1 = 0.08$, $w_2 = 0.22$, $w_3 = 0.32$, and $w_4 = 0.38$, consistent with Proposition 2 below. In this example the intersection-based set is visibly more asymmetric than the weight-based set. The weight-based ambiguity set has a symmetric shape and it can be shown that this is in fact a circle, with a vertical axis at the weighted average of the observations. More generally we can say that the ambiguity set for the weighted approach contains fewer distributions supported on a small interval (i.e., near the horizontal axis).

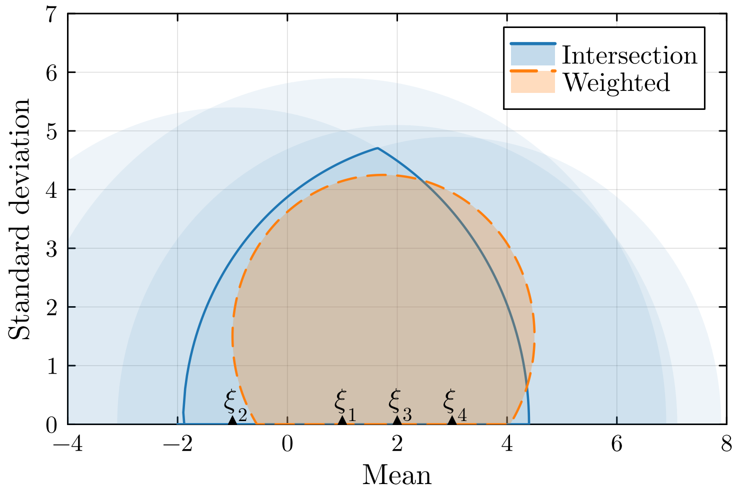

Figure 1: **Ambiguity Sets.** Means and standard deviations of uniform distributions under the weighted and intersection approaches for a one-dimensional example with $p = 2$

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3 Concentration of Measure for Weighted Empiricals

The by now classical results [7, Theorem 1 and Theorem 2] show that when $\Xi \subset \mathbb{R}^m$ is bounded and $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_N \sim \mathbb{P}_{T+1} \in \mathfrak{P}(\Xi)$ are independent and identically distributed random samples, then there exist constants $c_0, c_1, c_2 > 0$ and $q \in (0, 1/2]$ (depending only on m, $\operatorname{diam}(\Xi)$, and p) such that

$$\mathbb{E}\left[W_p^p\left(\tfrac{1}{N}\textstyle\sum_{i=1}^N \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}_{T+1}\right)\right] \le c_0 N^{-q} \quad \text{for all} \quad N \in \mathbb{N}, \tag{3}$$

and

$$\Pr\left[W_p\left(\tfrac{1}{N}\textstyle\sum_{i=1}^N \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}_{T+1}\right) \ge \varepsilon\right] \le c_1 \cdot \exp\left(-c_2 N \varepsilon^{p/q}\right) \quad \text{for all} \quad N \in \mathbb{N},\ \varepsilon \in \mathbb{R}_+. \tag{4}$$

Here $q \coloneqq \min\{p/m, 1/2\}$ when $p \neq m/2$. The boundary case $p = m/2$ entails a logarithmic correction; nevertheless, one can account for this while achieving (3) by choosing a q value slightly lower than $1/2$, which corresponds to a weaker bound in (3). Henceforth, for a given m and p, we take $q = \min\{p/m, 1/2\} - \delta$ for some small $\delta \ge 0$ so that $q \in (0, 1/2)$ achieves (3).

The goal of this section is to derive an analogue of (4) when the samples $\boldsymbol{\xi}_1 \sim \mathbb{P}_1, \ldots, \boldsymbol{\xi}_T \sim \mathbb{P}_T$ are independent but not identically distributed and when the equally weighted empirical distribution $\frac{1}{T}\sum_{t=1}^T \mathbb{1}_{\boldsymbol{\xi}_t}$ is replaced by a weighted one, $\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}$, for $w \coloneqq (w_1, \ldots, w_T) \in \mathcal{W}_T$. Our bound will explicitly reflect the drift between $\mathbb{P}_1, \ldots, \mathbb{P}_T$ and $\mathbb{P}_{T+1}$, and it will represent the weighting in the empirical distribution through the *effective sample size*

$$N_{\text{eff}}(w_1, \ldots, w_N) \coloneqq \frac{1}{\sum_{i=1}^N w_i^2}.$$

This approximates the number of equally weighted samples that would yield the same variance as the weighted sample [4, 12]. It ranges from 1 to N, with its extremes attained at $N_{\text{eff}}(w) = 1$ when all weight is placed on a single observation and $N_{\text{eff}}(w) = N$ for uniform weights $w_i = 1/N$.

Throughout this section, our indices and weights match the setting: we use $(i, N, N_{\text{eff}}, w_i, \mathcal{W}_N)$ for stationary results and $(t, T, T_{\text{eff}}, w_t, \mathcal{W}_T)$ for nonstationary results, respectively.

We derive our result in two stages. In the first stage we analyze the stationary case where the samples are drawn independently from the same distribution. The concentration bounds of Fournier & Guillin [7] are proved for samples with equal weights, and we wish to generalize to our setting. We begin from their expectation bound and in Lemma 1 extend this bound to arbitrarily weighted empirical distributions. Then Proposition 1 upgrades it to a high-probability concentration bound via a McDiarmid argument. In the second stage we turn to the nonstationary case where the samples are drawn from differing distributions. Lemma 2 shows how the Wasserstein distance between the weighted empirical distribution formed from these nonstationary samples and the next-period distribution $\mathbb{P}_{T+1}$ relates to the Wasserstein distance between a counterpart weighted empirical distribution formed from stationary samples and $\mathbb{P}_{T+1}$. Finally, Theorem 1 combines Proposition 1 with Lemma 2 to deliver our main concentration bound for nonstationary data.

Lemma 1. *For a distribution $\mathbb{P} \in \mathfrak{P}(\Xi)$ with bounded moments, let $q \in (0, 1/2)$ achieve* (3)*. Then, there exists a constant $c_0 > 0$ (depending only on m, $\mathbb{P}$'s moment bounds, p, and q) such that*

$$\mathbb{E}_{\mathbb{P}^N}\left[W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\xi_i}, \mathbb{P}\Big)\right] \leq c_0 N_{\text{eff}}(w)^{-q} \quad \textit{for all } N \in \mathbb{N},\ w \in \mathcal{W}_N.$$

Proof. Our proof proceeds in four steps. Step 1 decomposes the Wasserstein distance from the statement of the lemma into a sum of Wasserstein distances involving auxiliary unweighted empirical distributions. Step 2 applies a known bound for expected Wasserstein distances of unweighted empirical distributions to each of the Wasserstein distances from Step 1. We then split the resulting sum of bounds into two parts, and Steps 3 and 4 show that each part is dominated by the expectation bound from the statement of the lemma.

In view of Step 1, we re-index the weights so that $w_1 \geq \cdots \geq w_N$. This entails no loss of generality since re-indexing does not change $N_{\text{eff}}(w)$. The weighted empirical distribution satisfies

$$\begin{aligned}\sum_{i=1}^N w_i \mathbb{1}_{\xi_i} &= w_N \cdot \sum_{i=1}^N \mathbb{1}_{\xi_i} + (w_{N-1} - w_N) \cdot \sum_{i=1}^{N-1} \mathbb{1}_{\xi_i} + \cdots + (w_1 - w_2)\mathbb{1}_{\xi_1} \\ &= N w_N \cdot \frac{1}{N} \cdot \sum_{i=1}^N \mathbb{1}_{\xi_i} + (N-1)(w_{N-1} - w_N) \cdot \frac{1}{N-1} \cdot \sum_{i=1}^{N-1} \mathbb{1}_{\xi_i} + \cdots + (w_1 - w_2)\mathbb{1}_{\xi_1},\end{aligned}$$

where the last expression constitutes a convex combination of the distributions $\frac{1}{N}\sum_{i=1}^N \mathbb{1}_{\xi_i}$, $\frac{1}{N-1}\sum_{i=1}^{N-1} \mathbb{1}_{\xi_i}$, …, $\mathbb{1}_{\xi_1}$ with coefficients $N w_N$, $(N-1)(w_{N-1} - w_N)$, …, $(w_1 - w_2)$. Thus, by the convexity of $\mu \mapsto W_p^p(\mu, \mathbb{P})$ in its first argument (see, e.g., [21, Theorem 4.8]), we obtain

$$\begin{aligned}W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\xi_i}, \mathbb{P}\Big) \leq N w_N W_p^p\Big(\frac{1}{N}\sum_{i=1}^N \mathbb{1}_{\xi_i}, \mathbb{P}\Big) + (N-1)(w_{N-1} - w_N) W_p^p\Big(\frac{1}{N-1}\sum_{i=1}^{N-1} \mathbb{1}_{\xi_i}, \mathbb{P}\Big) \\ + \cdots + (w_1 - w_2) W_p^p\Big(\mathbb{1}_{\xi_1}, \mathbb{P}\Big).\end{aligned}$$

As for Step 2, we invoke the expectation bound of [7, Theorem 1] to each Wasserstein distance in the previous expression. This bound implies that there is a constant $c_0 > 0$ (depending only on m, $\mathbb{P}$'s moment bounds, and p) such that

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}^N}\left[W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big)\right] \\
&\leq N w_N c_0 N^{-q} + (N-1)(w_{N-1} - w_N) c_0 (N-1)^{-q} + \cdots + (w_1 - w_2) c_0 \\
&= N^{1-q} w_N c_0 + (N-1)^{1-q}(w_{N-1} - w_N) c_0 + \cdots + (w_1 - w_2) c_0 \\
&= \Big(N^{1-q} - (N-1)^{1-q}\Big) w_N c_0 + \Big((N-1)^{1-q} - (N-2)^{1-q}\Big) w_{N-1} c_0 + \cdots + w_1 c_0.
\end{aligned}$$

Thus, we obtain that

$$\mathbb{E}_{\mathbb{P}^N}\left[W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big)\right] \leq c_0 \cdot \sum_{i=1}^N \Big(i^{1-q} - (i-1)^{1-q}\Big) w_i \leq c_0 \cdot \sum_{i=1}^N i^{-q} w_i, \tag{5}$$

where the final inequality follows since $i^{1-q} - (i-1)^{1-q} \leq i^{-q}$ for all $i \in \mathbb{N}$ when $q > 0$. For the subsequent steps, in (5) we decompose the final expression $\sum_{i=1}^N i^{-q} w_i$ into $H + R$ with

$$H = \sum_{i=1}^{\lfloor N_{\text{eff}}(w) \rfloor} i^{-q} w_i \quad \text{and} \quad R = \sum_{i=\lfloor N_{\text{eff}}(w) \rfloor + 1}^{N} i^{-q} w_i.$$

Step 3 inspects the 'head' term H. By Cauchy–Schwarz and the definition of $N_{\text{eff}}(w)$,

$$H \leq \Big(\textstyle\sum_{i=1}^{\lfloor N_{\text{eff}}(w) \rfloor} i^{-2q}\Big)^{1/2} \cdot \Big(\textstyle\sum_{i=1}^{\lfloor N_{\text{eff}}(w) \rfloor} w_i^2\Big)^{1/2} \leq \Big(\textstyle\sum_{i=1}^{\lfloor N_{\text{eff}}(w) \rfloor} i^{-2q}\Big)^{1/2} \cdot N_{\text{eff}}(w)^{-1/2}.$$

Since $x \mapsto x^{-2q}$ is decreasing, we have

$$\sum_{i=1}^{\lfloor N_{\text{eff}}(w) \rfloor} i^{-2q} \leq 1 + \int_1^{\lfloor N_{\text{eff}}(w) \rfloor} x^{-2q} \, dx \leq \frac{\lfloor N_{\text{eff}}(w) \rfloor^{1-2q}}{1-2q},$$

and thus

$$H \leq \frac{1}{\sqrt{1-2q}} \cdot \lfloor N_{\text{eff}}(w) \rfloor^{1/2-q} \cdot N_{\text{eff}}(w)^{-1/2} \leq \frac{1}{\sqrt{1-2q}} \cdot N_{\text{eff}}(w)^{-q},$$

where the final inequality uses that $\lfloor N_{\text{eff}}(w) \rfloor \leq N_{\text{eff}}(w)$ and $1/2 - q > 0$.

Step 4, finally, bounds the 'remainder' term R. Since $i \mapsto i^{-q}$ is decreasing and the weights w_i sum to 1, we have

$$R \leq \sum_{i=\lfloor N_{\text{eff}}(w) \rfloor + 1}^{N} \Big(\lfloor N_{\text{eff}}(w) \rfloor + 1\Big)^{-q} \cdot w_i \leq \Big(\lfloor N_{\text{eff}}(w) \rfloor + 1\Big)^{-q} \leq N_{\text{eff}}(w)^{-q},$$

where the final inequality holds since $\lfloor N_{\text{eff}}(w) \rfloor + 1 \geq N_{\text{eff}}(w)$. Combining (5) with the bounds for H (Step 3) and R (above) completes the proof with $c_0' = c_0(1 + 1/\sqrt{1-2q})$. □

Lemma 1 generalizes [7, Theorem 1 case 3] to the setting of weighted empirical distributions. The convergence rates of both results coincide; the only difference is that their bound uses the actual sample size N, whereas ours uses the effective sample size $N_{\text{eff}}(w)$.

Under the additional assumption that Ξ is bounded, we can use the expectation bound of Lemma 1 to give a concentration inequality. Note that the boundedness of Ξ implies that every distribution $\mathbb{P} \in \mathfrak{P}(\Xi)$ has bounded moments.

Proposition 1 (Concentration with Effective Sample Size)**.** *Assume* $\operatorname{diam}(\Xi) < \infty$*. For a distribution $\mathbb{P} \in \mathfrak{P}(\Xi)$, let $q \in (0, 1/2)$ be the exponent from Lemma 1. Then for independent random samples $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_N \sim \mathbb{P}$, there exist constants $c_1 > 0$ (depending only on* $\operatorname{diam}(\Xi)$ *and p) and $c_2 > 0$ (from Lemma 1) such that*

$$\Pr\Big[W_p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big) \geq \varepsilon\Big] \leq \exp\Big(-c_1 N_{\text{eff}}(w) \cdot \Big(\varepsilon^p - c_2 N_{\text{eff}}(w)^{-q}\Big)_+^2\Big)$$
$$\text{for all } N \in \mathbb{N},\ w \in \mathcal{W}_N,\ \varepsilon \in \mathbb{R}_+. \quad (6)$$

In particular, whenever $\varepsilon \geq 2\big(c_2 N_{\text{eff}}(w)^{-q}\big)^{1/p}$, then

$$\Pr\Big[W_p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big) \geq \varepsilon\Big] \leq \exp\Big(-\frac{c_1}{4} \cdot N_{\text{eff}}(w)\varepsilon^{2p}\Big). \quad (6')$$

Proof. We prove the result by applying McDiarmid's inequality to the random variable $W_p^p(\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P})$. To this end, it suffices to show that $(\xi_1, \ldots, \xi_N) \mapsto W_p^p(\sum_{i=1}^N w_i \mathbb{1}_{\xi_i}, \mathbb{P})$ satisfies the bounded-differences condition, that is, to bound $\big|W_p^p(\sum_{i=1}^N w_i \mathbb{1}_{\xi_i}, \mathbb{P}) - W_p^p(\sum_{i=1}^N w_i \mathbb{1}_{\xi_i'}, \mathbb{P})\big|$ for all $(\xi_1, \ldots, \xi_N), (\xi_1', \ldots, \xi_N') \in \Xi^N$ with $\xi_i = \xi_i'$ for all $i \neq j$ and some $j \in [N]$.

Set $\mu' \coloneqq \sum_{i=1}^N w_i \mathbb{1}_{\xi_i'}$, and let $\gamma^{\star\prime}$ be an optimal coupling of μ' and $\mathbb{P}$, thus achieving cost $W_p^p(\sum_{i=1}^N w_i \mathbb{1}_{\xi_i'}, \mathbb{P})$. Since μ' is discrete, there exist conditional laws $\beta_i^{\star\prime} \in \mathfrak{P}(\Xi)$ such that

$$\gamma^{\star\prime}(\{\xi_i'\} \times \mathcal{B}) = w_i \beta_i^{\star\prime}(\mathcal{B}) \quad \text{for all measurable } \mathcal{B} \subseteq \Xi.$$

Define a coupling γ of $\mu \coloneqq \sum_{i=1}^N w_i \mathbb{1}_{\xi_i}$ and $\mathbb{P}$ by

$$\gamma(\{\xi_i\} \times \mathcal{B}) \coloneqq w_i \beta_i^{\star\prime}(\mathcal{B}) \quad \text{for all measurable } \mathcal{B} \subseteq \Xi \text{ and all } i \in [N].$$

Since $\xi_i = \xi_i'$ for all $i \neq j$, these slices coincide with those of $\gamma^{\star\prime}$; only the j^{th} slice is relocated from $\{\xi_j'\} \times \mathcal{B}$ to $\{\xi_j\} \times \mathcal{B}$ while keeping the same conditional $\beta_j^{\star\prime}$. With this disintegration, the cost of the optimal coupling $\gamma^{\star\prime}$ is

$$W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\xi_i'}, \mathbb{P}\Big) = \displaystyle\sum_{i=1}^N w_i \mathbb{E}_{\beta_i^{\star\prime}}\big[\|\xi_i' - \boldsymbol{\zeta}\|^p\big].$$

For the feasible coupling γ, we analogously obtain

$$W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\xi_i}, \mathbb{P}\Big) \leq \displaystyle\sum_{i \neq j} w_i \mathbb{E}_{\beta_i^{\star\prime}}\big[\|\xi_i' - \boldsymbol{\zeta}\|^p\big] + w_j \mathbb{E}_{\beta_j^{\star\prime}}\big[\|\xi_j - \boldsymbol{\zeta}\|^p\big].$$

Subtracting the two expressions yields

$$W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\xi_i}, \mathbb{P}\Big) - W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\xi_i'}, \mathbb{P}\Big) \le w_j \Big(\mathbb{E}_{\beta_j^{\star\prime}}\big[\|\xi_j - \boldsymbol{\zeta}\|^p\big] - \mathbb{E}_{\beta_j^{\star\prime}}\big[\|\xi_j' - \boldsymbol{\zeta}\|^p\big]\Big)$$
$$\le w_j \cdot \mathrm{diam}(\Xi)^p,$$

since $0 \le \|\xi - \zeta\|^p \le \mathrm{diam}(\Xi)^p$ for all $\xi, \zeta \in \Xi$. A symmetric argument then shows

$$\left| W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\xi_i}, \mathbb{P}\Big) - W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\xi_i'}, \mathbb{P}\Big) \right| \le w_j \cdot \mathrm{diam}(\Xi)^p.$$

Applying McDiarmid's inequality with the bounded-differences constants $a_i \coloneqq w_i \cdot \mathrm{diam}(\Xi)^p$, so that $\sum_{i=1}^N a_i^2 = \mathrm{diam}(\Xi)^{2p} \cdot \sum_{i=1}^N w_i^2$, we obtain

$$\Pr\Big[W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big) - \mathbb{E}_{\mathbb{P}^N}\Big[W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big)\Big] \ge \varepsilon\Big] \le \exp\Big(-c_1 N_{\mathrm{eff}}(w) \varepsilon^2\Big)$$
$$\text{for all } N \in \mathbb{N},\ w \in \mathcal{W}_N,\ \varepsilon \in \mathbb{R}_+,$$

where $c_1 = 2 \cdot \mathrm{diam}(\Xi)^{-2p}$. Using the expectation bound of Lemma 1, we obtain

$$\begin{aligned}
&\Pr\Big[W_p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big) \ge \varepsilon\Big] \\
&= \Pr\Big[W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big) \ge \varepsilon^p\Big] \\
&\le \Pr\Big[W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big) - \mathbb{E}_{\mathbb{P}^N}\Big[W_p^p\Big(\textstyle\sum_{i=1}^N w_i \mathbb{1}_{\boldsymbol{\xi}_i}, \mathbb{P}\Big)\Big] \ge \varepsilon^p - c_2 N_{\mathrm{eff}}(w)^{-q}\Big] \\
&\le \exp\Big(-c_1 N_{\mathrm{eff}}(w) \cdot \Big(\varepsilon^p - c_2 N_{\mathrm{eff}}(w)^{-q}\Big)_+^2\Big) \quad \text{for all } N \in \mathbb{N},\ w \in \mathcal{W}_N,\ \varepsilon \in \mathbb{R}_+,
\end{aligned}$$

which is (6). Finally, if $\varepsilon^p \ge 2c_2 N_{\mathrm{eff}}(w)^{-q}$, then $\big(\varepsilon^p - c_2 N_{\mathrm{eff}}(w)^{-q}\big)_+ \ge \varepsilon^p/2$, yielding (6'). □

Proposition 1 shows that the weighted empirical distribution concentrates around its mean at an exponential rate governed by the effective sample size $N_{\mathrm{eff}}(w)$. The simplified bound (6') highlights that once ε exceeds a constant multiple of the natural scale $N_{\mathrm{eff}}(w)^{-q/p}$, the tail probability decays at least as fast as $\exp(-\frac{c_1}{4} N_{\mathrm{eff}}(w)\varepsilon^{2p})$.

We now turn to the nonstationary case, where the samples $\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_T$ are drawn from differing distributions $\mathbb{P}_1, \dots, \mathbb{P}_T$. As discussed in Section 2, we assume that changes between consecutive distributions are bounded in Wasserstein distance. We will make use of the *normalized weighted drift*,

$$D_p(w) \coloneqq \left(\sum_{t=1}^T w_t (T - t + 1)^p\right)^{1/p},$$

which is the L_p norm of the look-back index $T - t + 1$ with the weights giving the probabilities of different values. It has a minimal value of 1 if $w_T = 1$ and other w values are zero.

Lemma 2. *Let $\mathbb{P}_1, \dots, \mathbb{P}_T, \mathbb{P}_{T+1} \in \mathfrak{P}(\Xi)$ satisfy $W_\infty(\mathbb{P}_t, \mathbb{P}_{t+1}) \le \rho$ for all $t \in [T]$. For any weights $w \in \mathcal{W}_T$ and independent random samples $\boldsymbol{\xi}_t \sim \mathbb{P}_t$, $\boldsymbol{\zeta}_t \sim \mathbb{P}_{T+1}$, there exists a coupling* *of the product distributions* $\prod_{t=1}^T \mathbb{P}_t$ *and* $\prod_{t=1}^T \mathbb{P}_{T+1}$ *under which*

$$W_p\Big(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\Big) \le W_p\Big(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\zeta}_t}, \mathbb{P}_{T+1}\Big) + D_p(w)\rho \quad \text{a.s.}$$

Proof. By the triangle inequality for Wasserstein distances (see, e.g., [21, page 94]), we have

$$\begin{aligned} W_p\Big(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\xi_t}, \mathbb{P}_{T+1}\Big) &\le W_p\Big(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\zeta_t}, \mathbb{P}_{T+1}\Big) + W_p\Big(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\xi_t}, \sum_{t=1}^T w_t \mathbb{1}_{\zeta_t}\Big) \\ &\le W_p\Big(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\zeta_t}, \mathbb{P}_{T+1}\Big) + \Big(\textstyle\sum_{t=1}^T w_t \|\xi_t - \zeta_t\|^p\Big)^{1/p}. \end{aligned} \tag{7}$$

The triangle inequality also implies that

$$W_\infty(\mathbb{P}_t, \mathbb{P}_{T+1}) \le W_\infty(\mathbb{P}_t, \mathbb{P}_{t+1}) + \cdots + W_\infty(\mathbb{P}_T, \mathbb{P}_{T+1}) \le (T-t+1)\rho \quad \text{for all } t \in [T].$$

Since the infimum in the definition of W_∞ is attained (see [8, Proposition 1]), for each $t \in [T]$, there exists a coupling $\gamma_t \in \Gamma(\mathbb{P}_t, \mathbb{P}_{T+1})$ such that

$$\|\boldsymbol{\xi}_t - \boldsymbol{\zeta}_t\| \le (T-t+1)\rho \quad \gamma_t\text{-a.s.}$$

The product of these couplings yields a joint coupling $\Upsilon \in \Gamma\big(\prod_{t=1}^T \mathbb{P}_t, \prod_{t=1}^T \mathbb{P}_{T+1}\big)$ under which

$$\sum_{t=1}^T w_t \|\boldsymbol{\xi}_t - \boldsymbol{\zeta}_t\|^p \le \sum_{t=1}^T w_t (T-t+1)^p \rho^p = D_p^p(w)\rho^p \quad \Upsilon\text{-a.s.}$$

Combining this with (7) proves the result. □

Lemma 2 couples the heterogeneous samples $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_T$ to an independent and identically distributed proxy $\boldsymbol{\zeta}_1, \ldots, \boldsymbol{\zeta}_T$ and shows that distances to the weighted empirical distribution differ from those to the proxy by at most the addition of the weighted drift $D_p(w)\rho$. The additive term $D_p(w)\rho$ is essentially sharp: one can readily construct simple sequences of distributions for which the inequality in Lemma 2 is tight.

We now combine the stationary concentration inequality from Proposition 1 with the drift bound of Lemma 2 to study the behavior of weighted empirical measures that are generated by nonstationary distributions.

Theorem 1 (Concentration for Drifting Distributions)**.** *Assume that* $\mathrm{diam}(\Xi) < \infty$. *Let* $\mathbb{P}_1, \ldots, \mathbb{P}_T, \mathbb{P}_{T+1} \in \mathfrak{P}(\Xi)$ *satisfy* $W_\infty(\mathbb{P}_t, \mathbb{P}_{t+1}) \le \rho$ *for all* $t \in [T]$, *and let* q *be the exponent from Lemma 1. For independent samples* $\boldsymbol{\xi}_t \sim \mathbb{P}_t$, *there exist constants* $c_1, c_2 > 0$ *(as in Proposition 1) such that*

$$\Pr\Big[W_p\Big(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\Big) \ge \varepsilon\Big] \le \exp\Big(-c_1 T_{\mathrm{eff}}(w) \cdot \Big((\varepsilon - D_p(w)\rho)_+^p - c_2 T_{\mathrm{eff}}(w)^{-q}\Big)_+^2\Big)$$
$$\textit{for all } T \in \mathbb{N},\ w \in \mathcal{W}_T,\ \varepsilon \in \mathbb{R}_+. \tag{8}$$

In particular, whenever $\varepsilon \geq D_p(w)\rho + 2(c_2 T_{\text{eff}}(w)^{-q})^{1/p}$, *then*

$$\Pr\left[W_p\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\right) \geq \varepsilon\right] \leq \exp\left(-\frac{c_1}{4} \cdot T_{\text{eff}}(w) \cdot \left(\varepsilon - D_p(w)\rho\right)_+^{2p}\right). \tag{8'}$$

Proof. For independent random samples $\boldsymbol{\xi}_1 \sim \mathbb{P}_1, \ldots, \boldsymbol{\xi}_T \sim \mathbb{P}_T$ and $\boldsymbol{\zeta}_1, \ldots, \boldsymbol{\zeta}_T \sim \mathbb{P}_{T+1}$, Lemma 2 provides a coupling Υ of the two T-product distributions under which

$$W_p\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\right) \leq W_p\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\zeta}_t}, \mathbb{P}_{T+1}\right) + D_p(w)\rho \quad \text{a.s.}$$

Hence,

$$\begin{aligned}
&\Pr\left[W_p\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\right) \geq \varepsilon\right] \\
&= \Upsilon\left[W_p\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\right) \geq \varepsilon\right] \\
&\leq \Upsilon\left[W_p\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\zeta}_t}, \mathbb{P}_{T+1}\right) \geq \left(\varepsilon - D_p(w)\rho\right)_+\right] \\
&= \Pr\left[W_p\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\zeta}_t}, \mathbb{P}_{T+1}\right) \geq \left(\varepsilon - D_p(w)\rho\right)_+\right] \quad \text{for all } T \in \mathbb{N},\ w \in \mathcal{W}_T,\ \varepsilon \in \mathbb{R}_+.
\end{aligned}$$

Applying Proposition 1 to the right-hand side yields (8). For the tail (8'), note that if $\varepsilon \geq D_p(w) + 2\left(c_2 N_{\text{eff}}(w)^{-q}\right)^{1/p}$, then

$$\left(\left(\varepsilon - D_p(w)\rho\right)^p - c_2 T_{\text{eff}}(w)^{-q}\right)_+ \geq \frac{1}{2} \cdot \left(\varepsilon - D_p(w)\rho\right)_+^p,$$

which gives the stated bound. □

Theorem 1 shows that, after accounting for the weighted drift $D_p(w)\rho$ and the stationary scale $T_{\text{eff}}(w)^{-q/p}$, the weighted empirical distribution concentrates about $\mathbb{P}_{T+1}$. In particular, once the radius ε exceeds $D_p(w)\rho$ by a constant multiple of $T_{\text{eff}}(w)^{-q/p}$, the probability of the event $W_p(\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}) \geq \varepsilon$ is bounded by $\exp(-\frac{c_2}{4} T_{\text{eff}}(w)(\varepsilon - D_p(w)\rho)_+^{2p})$. When $\rho = 0$ the weighted drift $D_p(w)\rho = 0$ and the bound reduces to that of Proposition 1.

4 Optimal Observation Weightings

Theorem 1 shows that, for a given ambiguity radius ε, different choices of observation weights w lead to different probabilistic guarantees that the next-period distribution $\mathbb{P}_{T+1}$ lies within Wasserstein distance ε of the weighted empirical distribution $\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}$. Conversely, for a fixed confidence level, different weightings require different radii to achieve the same guarantee. It is therefore natural to seek the weighting that attains a prescribed probabilistic guarantee at the smallest possible radius, thereby yielding the least conservative ambiguity set.

To make this problem tractable we focus on the tail regime of Theorem 1 where (8') applies.

This corresponds to moderate values of ε, and the bound simplifies to

$$\Pr\Big[W_p\Big(\textstyle\sum_{t=1}^{T} w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\Big) \geq \varepsilon\Big] \leq \exp\Big(-\frac{c_1}{4} \cdot T_{\text{eff}}(w) \cdot \Big(\varepsilon - D_p(w)\rho\Big)_+^{2p}\Big).$$

In this regime minimizing the conservatism of the ambiguity set – that is, minimizing the ε required to achieve a given confidence level – is equivalent to maximizing the magnitude of the exponent on the right-hand side. Accordingly, we formulate the problem of selecting the optimal weighting w as

$$\underset{w \in \mathcal{W}_T}{\text{maximize}} \quad T_{\text{eff}}(w) \cdot \Big(\varepsilon - D_p(w)\rho\Big)_+^{2p}. \tag{9}$$

This objective presents a fundamental trade-off between two opposing forces: on one hand, *reducing sampling variance* by increasing the effective sample size $T_{\text{eff}}(w)$; and on the other, *controlling distributional drift* by limiting the weighted drift $D_p(w)$. Putting more weight on recent samples reduces exposure to drift but decreases the number of effectively independent samples. Note that dividing the objective in (9) by ρ^{2p} implies that the optimal solutions depend solely on the ratio ε/ρ rather than on the individual values of ε and ρ.

The problem (9) is high dimensional and nonconvex, and thus appears to be computationally challenging. Fortunately, it admits a significant simplification: it always possesses an optimal solution that lies within a simple two-parameter family of weights which are monotone truncated polynomials in the look-back index.

Theorem 2 (Structure of Optimal Weights)**.** *The problem* (9) *admits an optimal solution*

$$w_t = \Big(c_1 - c_2(T - t + 1)^p\Big)_+, \quad t \in [T],$$

for some constants $c_1, c_2 \geq 0$.

Proof. The positive-part operation in (9) must be inactive at optimality; otherwise any feasible w would be optimal. We may thus assume $\varepsilon > D_p(w)\rho$ and equivalently maximize

$$T_{\text{eff}}(w)\Big(\varepsilon - D_p(w)\rho\Big)^{2p}$$

over $w \in \mathcal{W}_T = \big\{w \in \mathbb{R}^T : \sum_{t=1}^{T} w_t = 1,\, w_t \geq 0,\, t \in [T]\big\}$. Since the objective is continuous and the feasible region is compact, optimal solutions exist. Moreover, the Linear Independence Constraint Qualification (LICQ) holds at every feasible point: if $\mathcal{A} = \{t \in [T] : w_t = 0\}$ is the active set, then $\mathcal{A} \subsetneq [T]$ since at least one weight must be strictly positive, and thus the gradients of the active inequalities together with the equality-constraint gradient are linearly independent. Consequently, the Karush–Kuhn–Tucker (KKT) conditions are necessary for local (and hence global) optimality. We claim that any KKT point of the problem must exhibit the structure stated in the theorem.

Introducing a multiplier λ for the equality constraint and multipliers $\mu_t \geq 0$ for the inequalities in $\mathcal{W}_T$, the Lagrangian reads

$$L(w, \lambda, \mu) = T_{\text{eff}}(w) \left(\varepsilon - D_p(w)\rho\right)^{2p} + \lambda\Big(\sum_{t=1}^{T} w_t - 1\Big) + \sum_{t=1}^{T} \mu_t w_t.$$

By the definitions of $T_{\text{eff}}(w)$ and $D_p(w)$, we have

$$\frac{\partial T_{\text{eff}}(w)}{\partial w_s} = -2w_s\, T_{\text{eff}}(w)^2 \quad \text{and} \quad \frac{\partial D_p(w)}{\partial w_s} = \frac{(T-s+1)^p}{p}\, D_p(w)^{1-p}.$$

For strictly positive weights $w_s > 0$, complementary slackness implies $\mu_s = 0$, and hence the stationarity condition $\partial L(w, \lambda, \mu)/\partial w_s = 0$ reduces to

$$-2w_s T_{\text{eff}}(w)^2 \left(\varepsilon - D_p(w)\rho\right)^{2p} - 2\rho T_{\text{eff}}(w) \left(\varepsilon - D_p(w)\rho\right)^{2p-1} (T-s+1)^p D_p(w)^{1-p} + \lambda = 0. \tag{10}$$

Dividing through by $2T_{\text{eff}}(w)^2 (\varepsilon - D_p(w)\rho)^{2p-1}$ yields the affine relation

$$w_s = c_1 - c_2 (T-s+1)^p,$$

where

$$c_1 = \frac{\lambda}{2T_{\text{eff}}(w)^2 (\varepsilon - D_p(w)\rho)^{2p}} \quad \text{and} \quad c_2 = \frac{\rho D_p(w)^{1-p}}{T_{\text{eff}}(w)(\varepsilon - D_p(w)\rho)}. \tag{11}$$

For indices where this expression becomes negative, the complementary-slackness condition enforces $w_s = 0$. The constants $c_1, c_2 \geq 0$ are determined by the normalization constraint $\sum_{t=1}^{T} w_t = 1$ and by the truncation threshold beyond which the weights vanish. Hence, every locally optimal weighting takes the form

$$w_t = \left(c_1 - c_2(T-t+1)^p\right)_+, \quad t \in [T],$$

as claimed. □

Theorem 2 shows that the optimal weightings decay polynomially when looking backwards in time, and that they are truncated to zero beyond a finite horizon. As p increases, the polynomial decay sharpens. Choosing c_2 very small can keep the polynomial term small for small look-back times but, for large p, there comes a point at which this term starts to grow very sharply, and so the weights approach a constant c_1 over a window. Figure 2 shows how these optimal weightings behave in a specific example. This effect is explained by the geometry of the Wasserstein distances: the exponent p affects how severely large transportation distances are penalized. For small p, both small and large drifts between consecutive distributions contribute moderately to the overall cost, so assigning weight to older (more drifted) samples remains beneficial in terms of increasing the overall objective. As p grows, however, the transport cost grows increasingly steep, making even moderate drifts disproportionately expensive. Optimal solutions therefore concentrate weight on a recent subset of samples to avoid these heavy penalties, sharply downweighting older data beyond a certain point in time.

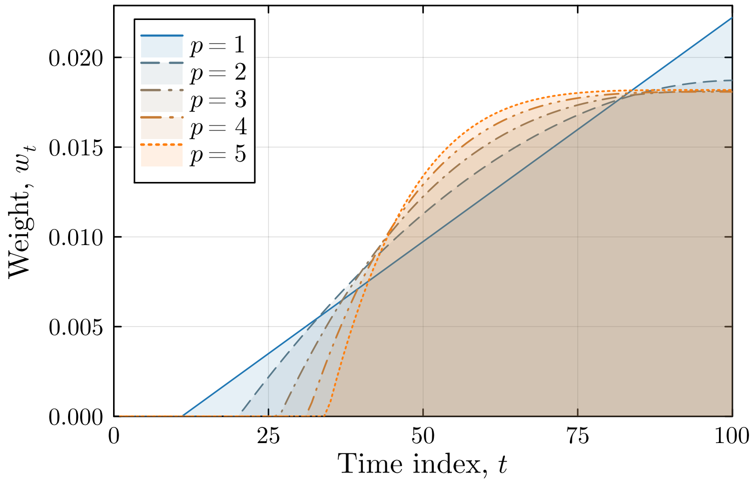

Figure 2: **Optimal Weightings for $p = 1, 2, 3, 4, 5$.** Here $\varepsilon/\rho = 90p$ and $T = 100$.

Theorem 2 also yields an efficient numerical solution method for (9). Without loss of generality, we may restrict ourselves to weightings of the form in Theorem 2 that are supported on the $S \in [T]$ most recent observations. For each fixed S, the simplex constraint $\mathcal{W}_S$ fixes $c_1 = (1 + c_2 \sum_{s=1}^{S} s^p)/S$, so the feasible set on that support collapses to a one-dimensional curve parameterized by c_2. Enforcing truncation exactly at S (i.e., $w_{T-S+1} > 0$ and $w_{T-S} = 0$), yields a single admissible interval $c_2 \in \big[1/(S(S+1)^p - \sum_{s=1}^{S} s^p), 1/(S^{p+1} - \sum_{s=1}^{S} s^p)\big)$. Thus, the global search for optimal weights reduces to a finite union of one-dimensional line searches.

The special case where $p = 1$ allows for a closed-form solution.

Proposition 2. *Assume $\varepsilon/\rho > 1$ and let $p = 1$. Then the problem* (9) *has an optimal solution supported on the $s = \lfloor \varepsilon/\rho \rfloor$ most recent observations given by*

$$w_t = \begin{cases} \dfrac{2((\varepsilon/\rho) + t - T - 1)}{s(2\varepsilon/\rho - s - 1)} & \text{if } t \geq T - s + 1, \\ 0 & \text{if } t \leq T - s. \end{cases} \tag{12}$$

Proof. Theorem 2 implies that any optimal solution to problem (9) with $p = 1$ has a truncated affine form, so that the weights are supported on a contiguous block of the most recent observations. Thus there exists an integer $s \in [T]$ such that

$$w_t = c_1 - c_2(T - t + 1) \quad \text{for } t \in \{T - s + 1, \ldots, T\}, \qquad w_t = 0 \text{ otherwise.}$$

Here the value of w_t at $t = T - s$ would be negative if it were not truncated. Hence $c_1 - c_2 s \geq 0$ and $c_1 - c_2(s + 1) < 0$, i.e., $s = \lfloor c_1/c_2 \rfloor$. Under this choice of s, we can calculate that

$$D_1(w) = \sum_{t=1}^{T} w_t(T - t + 1) = \sum_{t=T-s+1}^{T} w_t \frac{c_1 - w_t}{c_2} = \frac{c_1}{c_2} - \frac{1}{c_2} \sum_{t=1}^{T} w_t^2, \tag{13}$$

where the second equality replaces $(T - t + 1)$ with $(c_1 - w_t)/c_2$, and the third uses the fact that the w_t sum to 1.

From (11) in the proof of Theorem 2 we know that

$$c_2 = \frac{\rho}{T_{\text{eff}}(w)(\varepsilon - D_1(w)\rho)}.$$

Plugging in $T_{\text{eff}}(w) = \left(\sum_{t=1}^T w_t^2\right)^{-1}$ and the expression (13) for $D_1(w)$ and cross-multiplying, we obtain

$$c_2\Big(\varepsilon - \frac{c_1\rho}{c_2}\Big) + \rho\sum_{t=1}^{T} w_t^2 = \rho\sum_{t=1}^{T} w_t^2.$$

We thus deduce that $c_1/c_2 = \varepsilon/\rho$, giving the value $s = \lfloor c_1/c_2 \rfloor = \lfloor \varepsilon/\rho \rfloor$.

Since the weights on the last s periods all satisfy $w_t = c_1 - c_2(T-t+1)$ and $\sum_{t=1}^T w_t = 1$, we have $sc_1 - \frac{1}{2}c_2 s(s+1) = 1$. Then, substituting $c_1 = (\varepsilon/\rho)c_2$ into this normalization equation, we deduce that $c_2 s\big((\varepsilon/\rho) - (s+1)/2\big) = 1$, and hence

$$c_2 = \frac{2\rho}{s(2\varepsilon - \rho(s+1))}.$$

Substituting this expression and $c_1 = (\varepsilon/\rho)c_2$ into $w_t = c_1 - c_2(T-t+1)$ proves the claim. $\square$

Proposition 2 shows that for $p = 1$ the optimal weights increase linearly with recency over the last $\lceil \varepsilon/\rho \rceil - 1$ periods and vanish beyond that horizon.

We now consider two common heuristic schemes for assigning observation weights. The first is *windowing*, in which equal weights are assigned to the most recent s observations and older data are discarded. A window of size $s \in [T]$ therefore sets

$$w_T = \cdots = w_{T-s+1} = \frac{1}{s}, \quad w_{T-s} = \cdots = w_1 = 0.$$

The second is *smoothing*, in which weights decay geometrically backward in time at a fixed rate. Parameterized by a decay rate $\alpha \in [0,1]$, this sets

$$w_T = \alpha,\ w_{T-1} = \alpha(1-\alpha),\ w_{T-2} = \alpha(1-\alpha)^2,\ \ldots,$$

where we truncate to the available history length and scale the sum to 1.[2]

Both schemes are simple and widely used, but they can also be treated in our theoretical framework. For $p = 1$, one can determine their optimal parameter values and directly compare them to the closed-form optimum of Proposition 2. The following result (stated without proof) summarizes these parameter choices and their relationship with ε/ρ. Here, for any $x \in \mathbb{R}$ and interval $[a,b] \subset \mathbb{R}$, we write $x_{[a,b]}$ to denote the projection of x onto $[a,b]$.

Proposition 3. *Let $p = 1$. Then:*

(i) *The optimal window size for the problem* (9) *is either*

$$s = \left\lfloor \frac{1}{3}\Big(2\varepsilon/\rho - 1\Big) \right\rfloor_{[1,T]} \quad \textit{or} \quad s = \left\lceil \frac{1}{3}\Big(2\varepsilon/\rho - 1\Big) \right\rceil_{[1,T]}.$$

[2]For $\alpha = 0$, we take the limiting form as $\alpha \to 0$, corresponding to the equal weighting $w_T = \cdots = w_1 = 1/T$.

(ii) *As $T \to \infty$, the optimal decay rate for the problem* (9) *is*

$$\alpha = \left(\frac{3}{\varepsilon/\rho + 1}\right)_{\left[(\rho/\varepsilon)_{[0,1]},1\right]}.$$

Proposition 3(i) offers a natural interpretation. Ignoring the integer projection, the optimal window size grows approximately linearly with ε/ρ: doubling the admissible radius roughly doubles the optimal look-back horizon. Intuitively, larger ambiguity radii tolerate greater distributional drift, making it beneficial to incorporate older (less relevant) observations into the weighted empirical distribution.

The optimal weightings identified by Theorem 2 yield the smallest (least conservative) radius ε that ensures coverage of the next-period distribution $\mathbb{P}_{T+1}$ at a prescribed confidence level $1-\beta \in (0,1]$. In the stationary case ($\rho = 0$), this radius tends to zero as T increases. In contrast, when $\rho > 0$, the ambiguity radius ε cannot vanish – its size must reflect the drift. The next result quantifies how the minimal confidence radius scales with ρ (and β); for clarity we focus on the case $p = 1$.

Proposition 4 (Scaling of the Confidence Radius)**.** *Assume* $\operatorname{diam}(\Xi) < \infty$*. Let $p = 1$ and $\beta \in (0,1)$. There exist constants $c_1, c_2, c_3 > 0$ (independent of T, ρ, and β), such that if the history length $T \geq \left(\frac{12}{c_2}\rho^{-2}\log(1/\beta)\right)^{1/3}$, then, for optimal weightings $w \in \mathcal{W}_T$,*

$$\Pr\left[W_1\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\right) \geq \varepsilon(\beta,\rho)\right] \leq \beta,$$

where the minimal achievable confidence radius is at most

$$\varepsilon(\beta,\rho) = \begin{cases} c_1\Big(\rho\cdot\log(1/\beta)\Big)^{1/3} + c_2\Big(\rho^{2q}\cdot\log(1/\beta)^{-q}\Big)^{1/3} & \text{if } \rho < \rho_\star(\beta), \\ \rho + c_3\cdot\log(1/\beta)^{1/2} & \text{if } \rho \geq \rho_\star(\beta), \end{cases}$$

and where the threshold $\rho_\star(\beta) = (\frac{12}{c_2}\log(1/\beta))^{1/2}$.

Proof. We assume that $T \geq \left(\frac{12}{c_2}\rho^{-2}\log(1/\beta)\right)^{1/3}$. With $p = 1$, Theorem 1 yields

$$\Pr\left[W_1\left(\textstyle\sum_{t=1}^T w_t \mathbb{1}_{\boldsymbol{\xi}_t}, \mathbb{P}_{T+1}\right) \geq \varepsilon\right] \leq \exp\Big(-c_2 T_{\text{eff}}(w)\Big(\big(\varepsilon - D_1(w)\rho\big)_+ - c_1 T_{\text{eff}}(w)^{-q}\Big)_+^2\Big). \quad (14)$$

Our goal is to determine the smallest value of ε for which the exponential on the right-hand side of (14) is at most β. We use the triangular weights $w(s) \in \mathcal{W}_T$ supported on the last $s \in [T]$ observations, as specified in Proposition 2. A direct calculation gives

$$D_1(w(s)) = \frac{s+2}{3} \quad \text{and} \quad T_{\text{eff}}(w(s)) = \frac{3s(s+1)}{2(2s+1)} \geq \frac{3s}{4}.$$

The probability bound (14) is guaranteed to be at most β once

$$\Big(\varepsilon - D_1(w(s))\rho\Big)_+ - c_1 T_{\text{eff}}(w(s))^{-q} \;\geq\; \left(\frac{\log(1/\beta)}{c_2 T_{\text{eff}}(w(s))}\right)^{1/2}, \quad (15)$$

since in that case the exponent on the right-hand side of (14) does not exceed $-\log(1/\beta)$. Note that $D_1(w(s))$ and $T_{\text{eff}}(w(s))$ both increase with s. To obtain the smallest ε for which the inequality (15) holds, we choose s to balance these opposing effects. Define $\rho_\star(\beta) := (\frac{12}{c_2}\log(1/\beta))^{1/2}$ and analyze the two regimes $\rho < \rho_\star(\beta)$ and $\rho \geq \rho_\star(\beta)$ separately.

Case 1: $\rho < \rho_\star(\beta)$. Define

$$s_\star := \Big(\frac{12}{c_2}\rho^{-2}\log(1/\beta)\Big)^{1/3} \quad \text{and set} \quad s = \lceil s_\star \rceil.$$

Our earlier assumption on T ensures $s \in [T]$. Since $D_1(w(s)) \leq ((s_\star + 1) + 2)/3$, we have

$$D_1(w(s))\rho \leq \frac{s_\star}{3}\rho + \rho \leq \frac{4}{3}\Big(\frac{12}{c_2}\rho\log(1/\beta)\Big)^{1/3},$$

where the second inequality uses $\rho < \rho_\star(\beta)$. Moreover,

$$\Big(\frac{\log(1/\beta)}{c_2 T_{\text{eff}}(w(s))}\Big)^{1/2} \leq \Big(\frac{4\log(1/\beta)}{3c_2 s_\star}\Big)^{1/2} = \frac{2}{\sqrt{3}}\Big(\frac{\log(1/\beta)\rho}{\sqrt{12}c_2}\Big)^{1/3}.$$

Define

$$c_3 := \frac{4}{3}\Big(\frac{12}{c_2}\Big)^{1/3} + \frac{2}{\sqrt{3}}\Big(\frac{1}{\sqrt{12}c_2}\Big)^{1/3}, \qquad c_4 := c_1\Big(\frac{4}{3}\Big)^{q}\Big(\frac{12}{c_2}\Big)^{-q/3},$$

and set

$$\varepsilon(\beta,\rho) = c_3\Big(\rho\log(1/\beta)\Big)^{1/3} + c_4\Big(\rho^{2q}\log(1/\beta)^{-q}\Big)^{1/3}.$$

Then

$$\varepsilon(\beta,\rho) - D_1(w(s))\rho \geq \Big(\frac{\log(1/\beta)}{c_2 T_{\text{eff}}(w(s))}\Big)^{1/2} \quad \text{and} \quad c_4\Big(\rho^{2q}\log(1/\beta)^{-q}\Big)^{1/3} \geq c_1 T_{\text{eff}}(w(s))^{-q},$$

using $T_{\text{eff}}(w(s)) \geq 3s_\star/4$. We conclude that this choice of $\varepsilon(\beta,\rho)$ and s satisfies (15), and therefore the right-hand side of (14) is bounded above by β.

Case 2: $\rho \geq \rho_\star(\beta)$. We set $s = 1$ so that $D_1(w) = 1$ and $T_{\text{eff}}(w(s)) = 1$, and with

$$\varepsilon(\beta,\rho) = \rho + \Big(\frac{\log(1/\beta)}{c_2}\Big)^{1/2} + c_1,$$

the inequality (15) holds as an equality, and hence the right-hand side of (14) equals β. □

Since $q \in (0, 1/2)$, Proposition 4 implies that, for small drift, the minimal confidence radius scales as at most $\rho^{1/3}$. For large drift, the optimal weighting strategy is to place all weight on the most recent observation, the drift dominates, and the minimal confidence radius scales as ρ. For comparison, using the intersection-of-balls approach, a union-bound argument shows that the radius of the t^{th} ball must satisfy a lower bound of the form $(T - t + 1)\rho + c_1$ to achieve $1 - \beta$ confidence (see also [17, Proposition 4]). This lower bound scales linearly in both T and ρ, whereas our weighted scheme achieves a radius that is effectively independent of T and scales as $\rho^{1/3}$ in the small-drift regime, becoming linear in ρ only when the drift is large.

5 Numerical Experiments

In this section we study a decision problem faced by a newsvendor supplying a pool of consumers with demand that evolves over time. We compare the weighted empirical–based approach previously developed in this paper to the intersection-based approach of [17], as well as to the ambiguity free approaches of SAA and smoothing. To understand the optimal expected performance of each approach, we assess performance ex post.

Consider a newsvendor with underage cost $c_\mathrm{u} > 0$ and overage cost $c_\mathrm{o} > 0$ supplying a random demand $\boldsymbol{\xi} \sim \mathbb{P} \in \mathfrak{P}(\Xi)$ where $\Xi \subset \mathbb{R}$. The optimal order quantity solves

$$\underset{x \in \Xi}{\text{minimize}} \quad \mathbb{E}_\mathbb{P}\Big[c_\mathrm{u}(\boldsymbol{\xi} - x)_+ + c_\mathrm{o}(x - \boldsymbol{\xi})_+\Big].$$

For an ambiguity set of distributions $\mathcal{P} \subset \mathfrak{P}(\Xi)$, the DRO order quantity instead solves

$$\underset{x \in \Xi}{\text{minimize}} \quad \sup_{\mathbb{Q} \in \mathcal{P}} \mathbb{E}_\mathbb{Q}\Big[c_\mathrm{u}(\boldsymbol{\xi} - x)_+ + c_\mathrm{o}(x - \boldsymbol{\xi})_+\Big].$$

Here $\mathcal{P}$ may be a Wasserstein ball centered around a weighted empirical distribution, or an intersection of multiple Wasserstein balls each centered around their own empirical distributions. Since the newsvendor cost function can be expressed as the pointwise maximum of a finite number of affine functions, in the aforementioned cases the DRO problem can be reformulated as a convex-conic program following [5, Corollary 5.1(i)] or [17, Corollary 2]. In what follows we use $p = 2$ for the Wasserstein ball–based ambiguity sets; $p = 1$ on the other hand would lead to a somewhat degenerate form of distributional robustness; see [5, Remark 6.7].

In our experiments we set $c_\mathrm{u} = 4$ and $c_\mathrm{o} = 1$. For the demand process, at each time t we set $\boldsymbol{\xi} \sim \text{Binomial}(1000, \theta_t)$, where $\theta_t \in [0, 1]$ is a time-varying parameter. This amounts to an economy of 1000 consumers who each independently demand 1 unit with probability θ_t. The parameter θ_t evolves over time from an initial value $\theta_1 = 1/3$ following a random walk with $\text{Uniform}(-\delta, \delta)$ innovations. (We project θ_t onto the interval $[0, 1]$ as necessary to ensure that it remains a valid probability.) The value of $\delta \geq 0$ therefore parameterizes the extent of underlying nonstationarity. Note that that this setup meets the conditions of the generalized concentration inequalities of Section 3, with the W_∞ drift bound determined by δ.

To compare the ex-post expected performance of each approach, we average over 100 simulations. In each simulation, for $T = 70$ historical demand observations, newsvendor orders are made using all previous demand observations (weighted or intersected appropriately), with a cost incurred following the next demand distribution realization at time $T = 71$. When computing this cost, we use an analytic formula for the expected newsvendor loss under Binomial demand, and a further average of 1000 jumps from time $T = 70$ to 71. For each approach, this cost is computed over a range of parameter values, the minimum among these being the ex-post optimal expected performance of that approach. Note that for the intersection approach, when the historical observations are such that the resulting intersection is empty, we double the constituent ball radii until it is nonempty; see [17, Proposition 4].

Table 1 presents the parameter ranges used for each approach. Here for scalars $a \leq b$ and a positive integer n, we use the notation $\texttt{LinRange}(a, b; n)$ for the set of n values forming an equally spaced sequence between (inclusive of) a and b. Further, for scalars $a \leq b \in (0, \infty)$ and a positive integer n, we use the notation $\texttt{LogRange}(a, b; n)$ for the set of n values forming a geometric sequence between (inclusive of) a and b. This turns out to be the natural choice for parameterizing both the smoothing decay rate α and the ratio ρ/ε.

Table 1: **Parameter Ranges**

Decay rate	$\alpha \in \{0\} \cup \texttt{LogRange}(10^{-4}, 1; 30)$
Ambiguity radius	$\varepsilon \in \{0\} \bigcup_{i=-1}^{1} \texttt{LinRange}(10^{i}, 10^{i+1}; 10)$
Drift bound	$\rho : \rho/\varepsilon \in \{0\} \cup \texttt{LogRange}(10^{-4}, 1; 30)$

We formulate the problem (5) as a convex-conic program in `Julia` [1] using `JuMP.jl` [15] and `MathOptInterface.jl` [13], and solve it with `Gurobi` [10]. Figure 3 graphs the expected performance of each method for different extents of underlying nonstationarity.

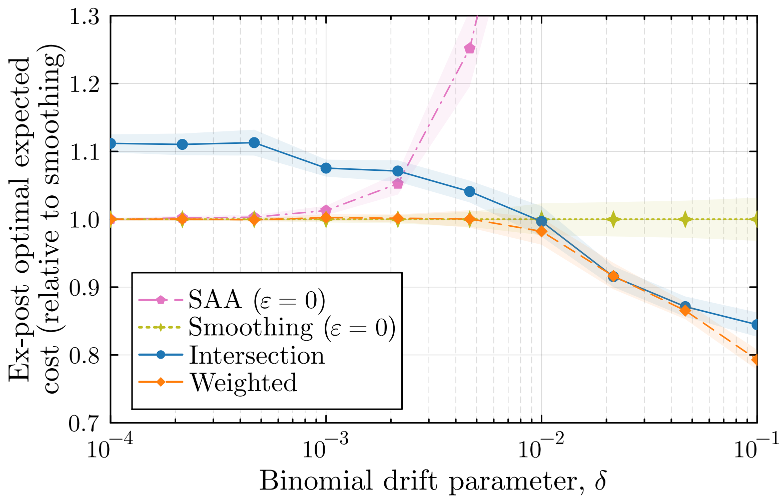

Figure 3: **Expected Performance.** Bands present standard-error ranges.

For a very small extent of nonstationary (essentially the stationary case), SAA, smoothing, and our weighted empirical–based approach all perform equally well, with the intersection-based approach performing $\approx 10\,\%$ worse. In view of Example 1, this is unsurprising. As the extent of nonstationarity increases, SAA quickly becomes the worst choice, and eventually, the two DRO approaches dominate. For intermediate to larger extents of nonstationarity, these approaches perform quite similarly, reflecting the drift threshold of Proposition 4 being crossed and the linear growth of each ambiguity ball. This holds up until the largest Binomial drift parameter, where the history length T compromises the intersection-based approach which performs $\approx 6\,\%$ worse than the weight empricial–based approach. (Do note however that Proposition 4 is proved in the different context of $p = 1$.) In any case, under all extents of nonstationarity, our weighted approach performs optimally up to statistical error.

6 Conclusions

This paper develops a distributionally robust optimization framework that explicitly accounts for temporal drift in the data-generating distribution. By centering Wasserstein ambiguity sets at weighted empirical distributions and deriving new concentration inequalities, we provide a principled way to trade off statistical variance against nonstationarity. The resulting optimal weightings have a simple polynomial–truncation structure that naturally generalizes widely used heuristic schemes such as exponential smoothing and windowing, and lead to tight confidence guarantees for the next-period distribution.

Our theoretical results yield clear insights into how the optimal weighting adapts to the magnitude of drift: when drift is small, long histories can be used effectively, whereas when drift is large, weight concentrates on the most recent observations. The numerical experiments confirm that the proposed weighting strategy matches or outperforms classical approaches across a wide range of nonstationarity levels.

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